## POWER: DC jack, SS14, VB, AMS1117-5.0 x2

## INPUT BUFFER (U3A) + OUTPUT MIXER (U3B): as Loop Echo / Small Time

## DELAY LINE 1: PT2399 (U1), network as Loop Echo / Small Time. TIME 1 = 850k + 2k7

## FEEDBACK (FB 1 / FB 2) -> limiter -> U4 buffers -> TWIST (dual gang): CCW = parallel, CW = Moebius

FB MAX trims: 0 = full feedback (as v5.2); turn up to lower the maximum repeat level.

Limiter: the loop gain can reach 1 with FB up (and stays 1 across TWIST), so the 1N4148s cap self-oscillation at about +-0.6 V.

## MODULATION: LFO (U6) -> DEPTH trim -> line 2 VCO only. RATE / DEPTH are internal trims

~0.5–12 Hz, injected at the TIME 2 / 2k7 node: the same DEPTH gives the same percentage wobble at any TIME setting.

## DELAY LINE 2: PT2399 (U2), network as Loop Echo / Small Time. TIME 2 = 850k + 2k7

Line 2 input: DRY (parallel) or W1 (series), set by SW1 SERIES. Line 2 VCO also gets the LFO (chorus-like spread).

## SERIES (SW1) + WET SUM (U5A) -> MIX. USB spare (future stereo R)

## JACKS (board-mounted, top wall), FOOTSWITCH HEADER (SwitchPCB v2), STATUS LED

Panel (1590BB landscape): TIME 1, TIME 2, MIX over FB 1, TWIST, FB 2; SERIES toggle; status LED; footswitch. Internal trims: RATE, DEPTH, FB MAX 1 / 2.

Blocks from Loop Echo (PT2399 network from Small Time). LFO on line 2, SMD for JLC (all basic parts).
Two PT2399 lines, TWIST blends self / cross feedback (parallel — Moebius), SERIES toggle

**Super-Freq**

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**Title: Dual Delay**

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